

Apply filters to SQL queries

Project description

This project is from the Linux and SQL module of the Google Cybersecurity Certificate. The goal here is to investigate a few potential security issues that involve login attempts and employee machines. Within SQL we will use filters to retrieve records from the pretend organization in the employees and log_in_attempts tables. The sections below will cover the commands used to run SQL queries and retrieve information from the database.

Retrieve after hours failed login attempts

We will examine failed login attempts after 18:00.

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00' AND success = 0;
```

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_time > '18:00' AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0

Retrieve login attempts on specific date

To examine login attempts on the 8th or 9th.

```
SELECT *  
FROM log_in_attempts  
WHERE login_time = '2022-05-09' OR login_time = '2022-05-08';
```

Retrieve login attempts outside of Mexico

Using NOT to exclude Mexico from the search instead of listing out all of the individual places. Easier this way.

```
SELECT *
```

```
FROM log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	0
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	0

Retrieve employees in Marketing

```
SELECT *
FROM employees
WHERE department = 'Marketing' AND office LIKE 'East-%';
```

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Marketing' AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267

Retrieve employees in Finance or Sales

```
SELECT *
FROM employees
WHERE department = 'Sales' OR department = 'Finance';
```

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170

Retrieve all employees not in IT

```
SELECT *  
FROM employees  
WHERE NOT department = 'Information Technology';
```

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434

Summary

- Investigated login activity by finding failed attempts after 18:00, identifying logins on May 8–9, 2022, and excluding any from Mexico.
- Analyzed employee data by listing Marketing staff in East offices, locating Finance or Sales staff, and showing all employees outside the IT department.